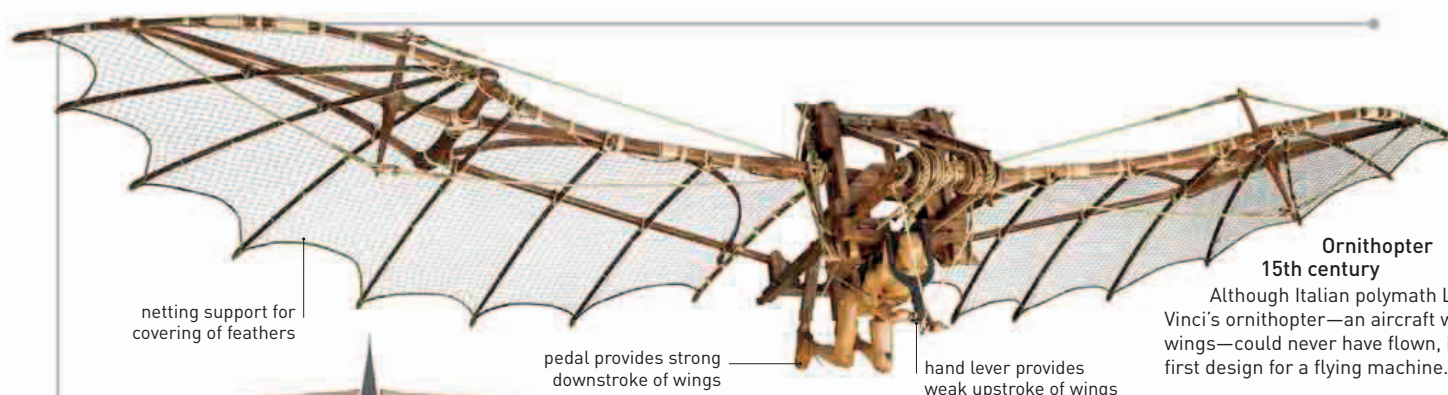




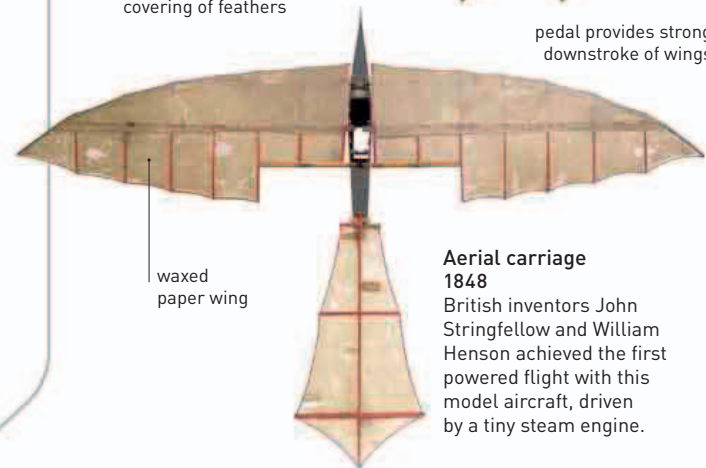
Ornithopter
15th century

Although Italian polymath Leonardo da Vinci's ornithopter—an aircraft with flapping wings—could never have flown, it may be the first design for a flying machine.

netting support for covering of feathers

pedal provides strong downstroke of wings

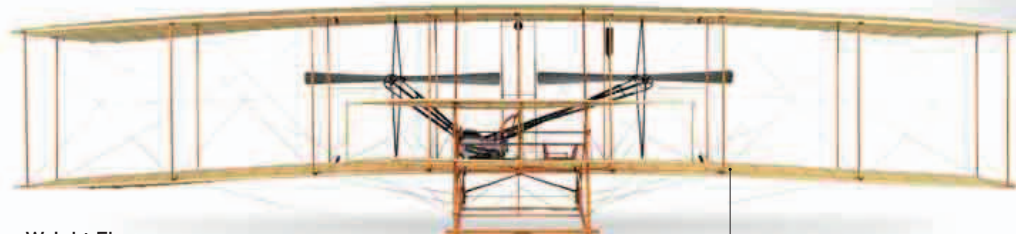
hand lever provides weak upstroke of wings



waxed paper wing

Aerial carriage
1848

British inventors John Stringfellow and William Henson achieved the first powered flight with this model aircraft, driven by a tiny steam engine.



Wright Flyer
1903

Orville and Wilbur Wright, American inventors, achieved the first manned, powered, controlled take-off and flight with their Flyer in North Carolina on December 17, 1903.

flexible wing helps control height and direction

FLYING MACHINES

THE DEVELOPMENT OF POWERED FLYING MACHINES LED TO A MILITARY AND TRANSPORT REVOLUTION

The dream of flight dates back to the Ancient Greek myth of Daedalus, who flew with wings of feather and wax. It was not until the 18th century that flight became a reality, paving the way for a range of flying machines.

Humans first took to the air in 1783, in a hot-air balloon developed by the Montgolfier brothers in France. For a century, aircraft used gas for lift—first in the form of balloons and then steerable airships. Winged aircraft took off under their own power for the first time with the Wright brothers' historic flight in 1903.



Hot-air balloon
Date unknown

Balloons gain lift because they contain a gas—such as helium or heated air—that is less dense than the air surrounding the balloon.



Skyship 500 HL
1984

Airships were the luxury liners of the 1920s, despite their slow speed and the inflammability of their lifting gas—hydrogen. Today, smaller airships serve as airborne platforms for cameras covering major events.

envelope filled with helium

gondola carries crew and passengers

tail rotor allows steering and stability



Schweizer 300C helicopter
1970

Small helicopters such as this one can hover, take off and land vertically, and turn on the spot—making them perfect for urban flying.

main rotor blades provide power and lift



tailfin 63 ft (19.3 m) high